Miguel Cruces Sánchez

Cacheable Oblivious DNS over HTTPS

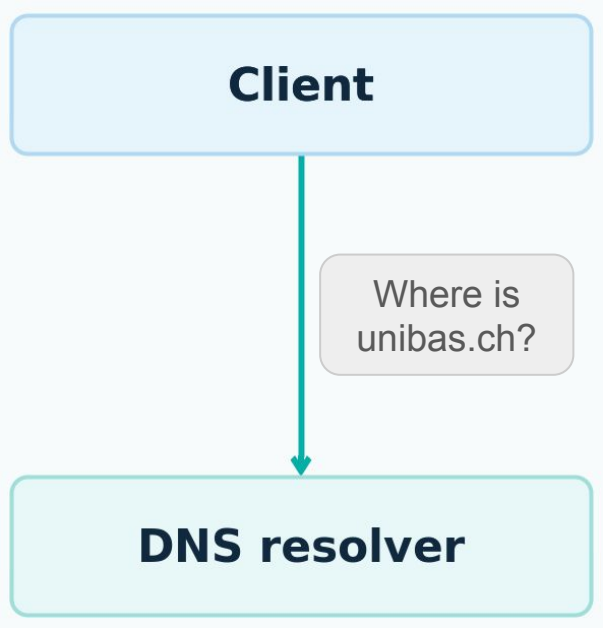
CODoH: Privacy-Preserving Caching

Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137

DNS lookups reveal a lot

Usually, opening a website starts with a **DNS query**.

DNS Queries **expose activity** and are **not encrypted**.



Generated by ChatGPT, modified by author

Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137

Outline

1. **DNS over HTTPS and Oblivious DoH**
2. **Cacheable ODoH**
3. **Cache Privacy Risks**
4. **Performance & Tradeoffs**

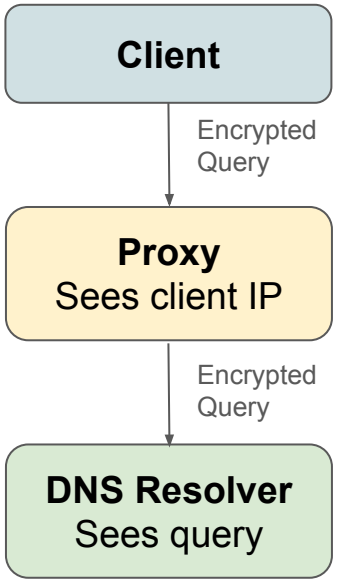
Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137

DoH and **ODoH** protect different information

DNS over HTTPS (DoH)

- Protects **from intermediaries**.
- DNS Resolver can associate the client's **IP address** and **plaintext query**.

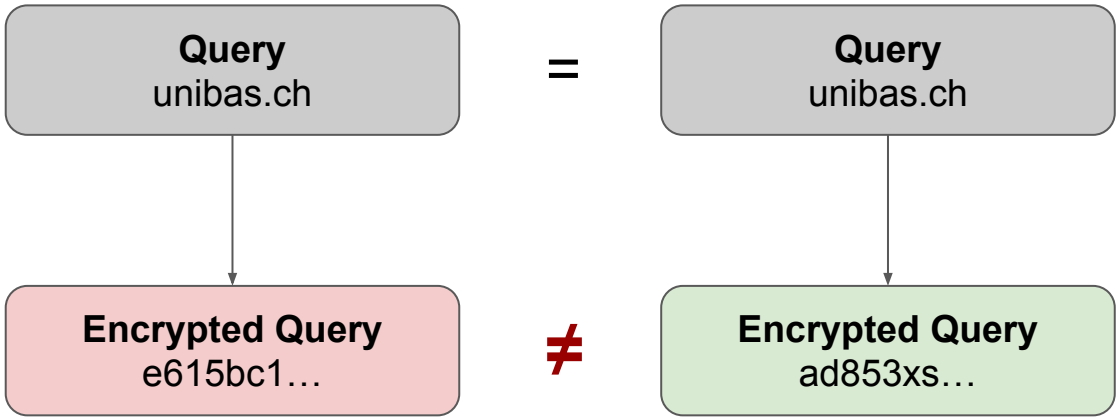
Oblivious DoH (ODoH)



Niroula et al., "CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS," PoPETs 2026, DOI: 10.56553/popets-2026-0137

ODoH breaks **proxy-level cache**

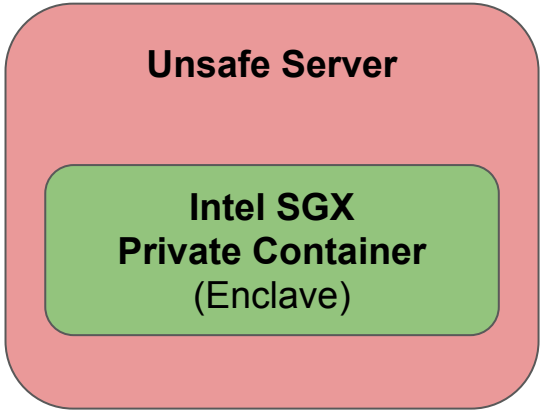
The proxy **can not tell** if two requests **ask for the same domain**.



Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137

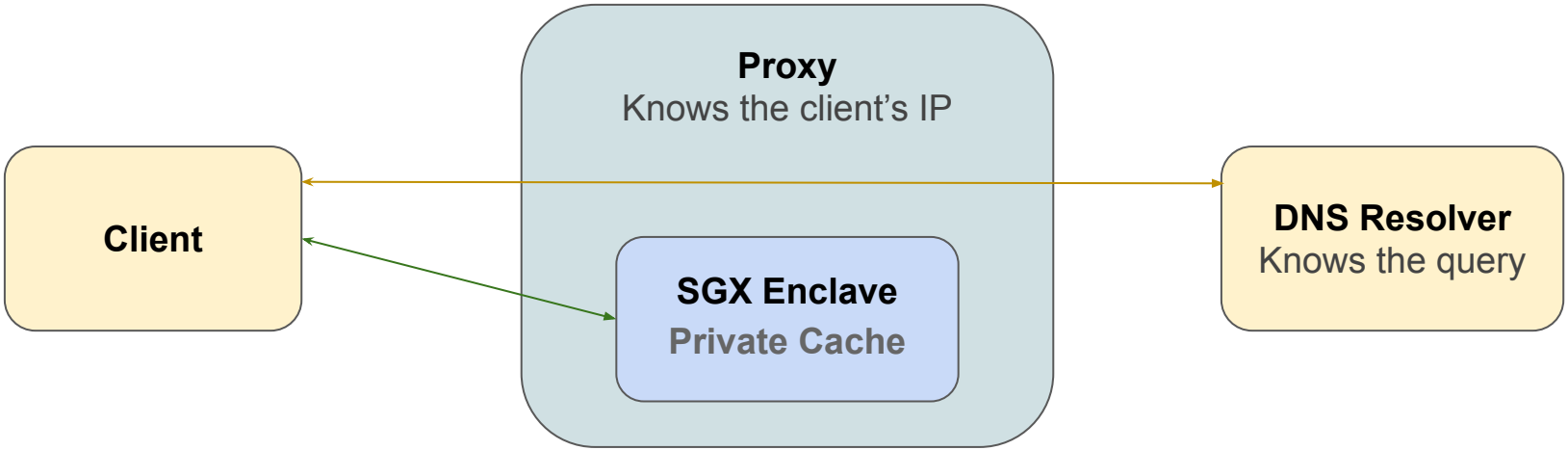
Intel SGX and Remote Attestation

- **Intel SGX** provides **safe, encrypted containers** called **enclaves**.
- **Remote attestation** proves **remote clients** that the expected code is running in a **real SGX enclave**.



So, how can we cache ODoH at proxies?

With **independently encrypted** queries.



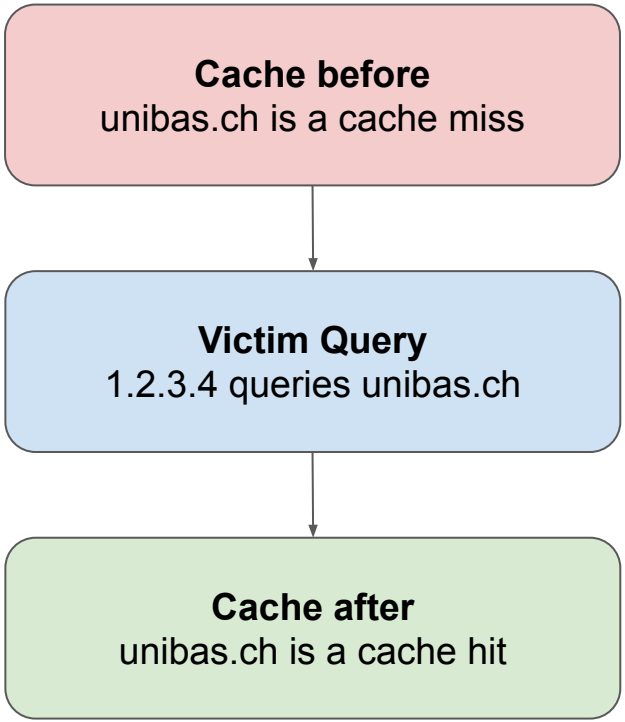
Niroula et al., "CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS," PoPETs 2026, DOI: 10.56553/popets-2026-0137

Caching creates new privacy risks

Set-difference attacks

Isolating a single query by **comparing cache states**.

A **malicious proxy** can infer **who** queried **what**.



Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137

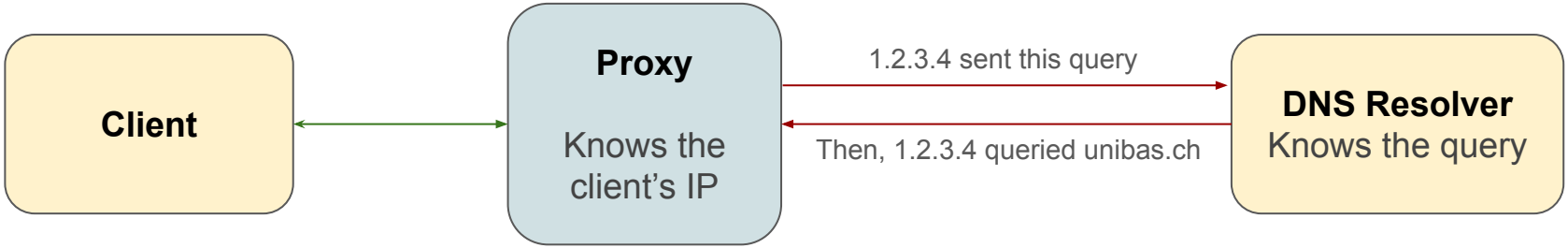
CODoH hides cache behavior

- **Batched updates** for cache inserts. No isolated updates.
- **Extra records** so queries are harder to isolate.
- **Oblivious RAM**. Same resource in different physical place every access.

Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137

It is private, but…

Like ODoH, depends on proxy and target **not colluding**.



Niroula et al., "CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS," PoPETs 2026, DOI: 10.56553/popets-2026-0137

Does it improve ODoH's performance?

Cold

10,000 queries
10,000 domains

All cache **misses**.

Zipf

10,000 queries
1,000 domains

Cache **hits** and **misses**.

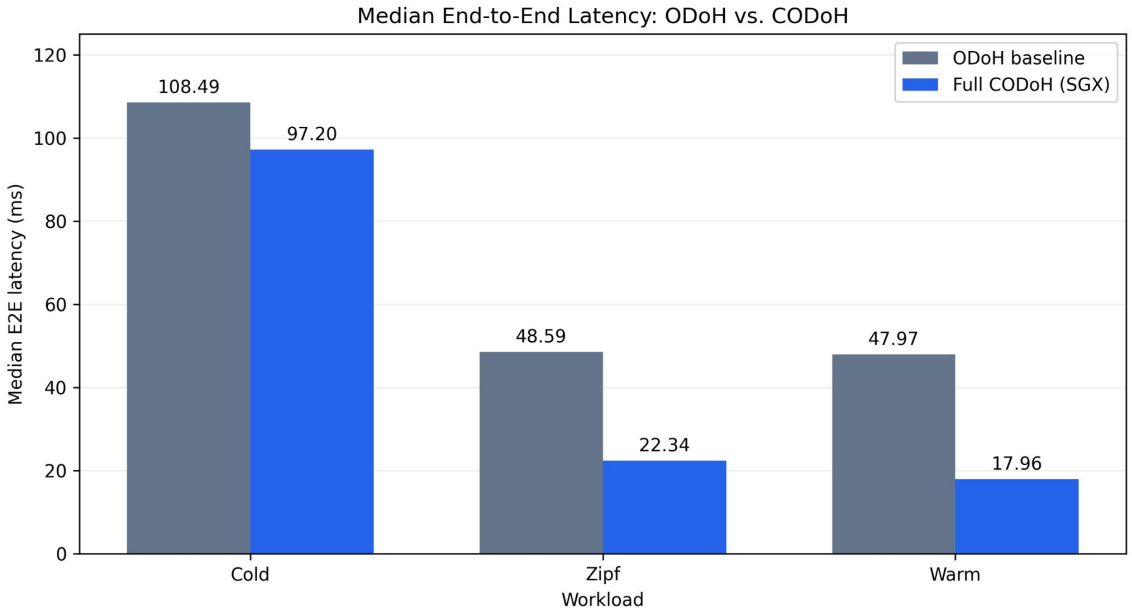
Warm

10,000 queries
1 domain

Cache **hits**.

Niroula et al., "CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS," PoPETs 2026, DOI: 10.56553/popets-2026-0137

Does it improve ODoH's performance? **Yes, up to 2.7x**



Adapted from Table 3 in Niroula et al. (2026). Source code generated with ChatGPT.

Better performance, but…

Low batch size (B=10) used for insertion in end to end benchmarks.

- A single page can do **related DNS requests**.
- Users **revisit** the **same pages** day after day.

⇒ Attackers can infer **sets of candidate visited websites**.

Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137

Planned programming project

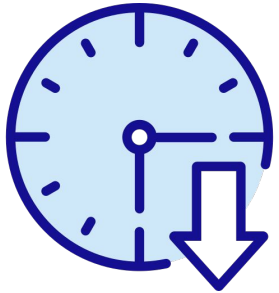
- Replicate **end to end benchmarks** with the paper's provided artifacts in **IBM Cloud**, using **3 VMs** in **different regions**.

- Explore **different batch sizes** and their **impact on latency**, taking **baseline B=10** (as in the paper)



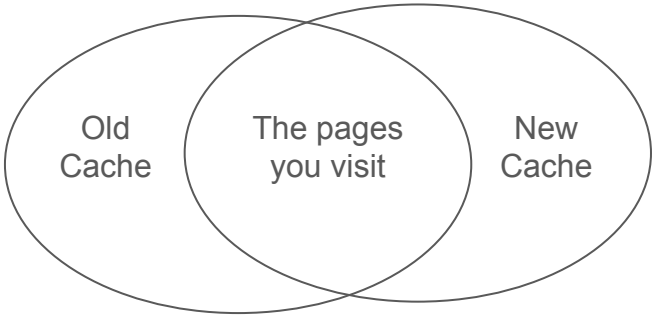


Conclusions



- Adding **cache** to **ODoH** results in **significant latency improvements**.

- Cache **introduces privacy risks** such as **set-difference attacks**.



Niroula et al., “CODoH: Privacy-Preserving Caching for Oblivious DNS over HTTPS,” PoPETs 2026, DOI: 10.56553/popets-2026-0137